Pro Team Multimodel — In a Nutshell

A Short Guide to the Interactive Form S-6 Grid, Wind Fields, Sensitivity, and Loss

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Abstract

This is the quick read. The **Pro Team Multimodel** is a zero-backend web application that draws the official Form S-6 21×40 grid over southern Florida, runs three hurricane wind-field models over it, turns wind into loss through a damage curve and an exposure book, and reports sensitivity, uncertainty, and loss metrics—the five catastrophe-model disciplines (*meteorology*, *vulnerability*, *statistics*, *actuarial science*, *computer science*) each made swappable or explorable. The full derivations, validations, and references are in the companion report; this document gives the essentials in one sitting. The single most important finding: once the storm population is sampled realistically—one lumped set of 200 storms carrying the observed dependence between hurricane parameters, rather than stratified into intensity categories—**central pressure emerges as the dominant driver of the loss**, a result the conventional per-category framing hides.

1 Summary

A catastrophe model is a chain: *exposure* (what is at risk and where) → *hazard* (the wind field) → *vulnerability* (how much damage a given wind does) → *loss* (dollars), with *statistics* quantifying how sensitive the answer is to each assumption. This tool implements that chain over a fixed experiment: a storm crosses a 60×117-mile grid in South Florida on a straight due-west track, and at each of the 840 grid vertices the app records the peak wind and the resulting loss. Everything downstream—sensitivity rankings, loss-cost distributions, financial exceedance curves—is computed from that per-vertex field.

2 Meteorology: grid, track, and wind fields

The grid and track. The lattice ($21 \times 40 = 840$ vertices at $\sim$3 statute-mile spacing) and its landfall point (25.86°N, 80.12°W) are *specified* by Form S-6 of the Florida Commission's Report of Activities [1]. The storm center starts 9 mi east of landfall and moves due west across the domain. Of the 840 vertices, 682 are land and 158 water. Placement maximizes *loss*—hazard × exposure—over the Miami–Fort Lauderdale corridor, not raw hurricane frequency (the frequency maximum lies south, over the sparsely-settled Keys).

Three wind models, plus a dynamic Powell. Each is evaluated at all 840 vertices, with translation asymmetry from the storm motion:

- **Powell** [5] — an iterative boundary-layer (slab) PDE solver; the physical reference.

- **Holland** [3] — the closed-form gradient-wind profile.
- **Willoughby** [4] — a piecewise power-law profile.

A fourth option, **Powell (dynamic)**, marches the slab PDE through physical time rather than solving a steady field; it matters only at the coastal transition, where it represents a lagged decay and an in-PDE drag that the steady model structurally cannot [6, 7]. The per-vertex metric is the **peak surface wind** over the storm's $-12 \to +24\,\mathrm{h}$ passage, sampled finely (1-minute steps) so the fast westward motion is not aliased. All footprint fields are *precomputed* offline and sampled instantly in the browser (Figure 1).

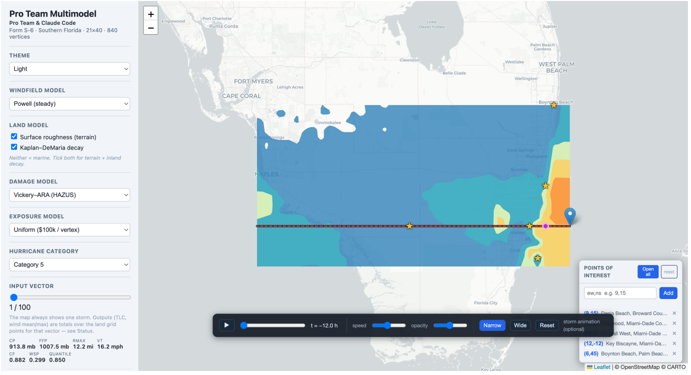

Figure 1: Powell peak-wind footprint (filled contours) for an intense example storm. Strongest winds lie north of the track—correct for a westward-moving Northern-Hemisphere cyclone, where translation adds to rotation on the north side.

Land effect: roughness and decay. Two independent, combinable toggles model the storm's interaction with land: **surface roughness** (a terrain-drag reduction from NLCD land cover [9] through published roughness tables [10, 11], land-mean factor ≈ 0.67) and **Kaplan–DeMaria decay** [8] (inland weakening with a Gulf-recovery term). Both are on by default; turning both off gives the marine field. For an intense storm the land-mean surface wind roughly halves from marine to fully-land-adjusted.

3 The storm population: a constrained, lumped design

This is the conceptual heart of the current tool, and where it departs from the ROA's category-stratified convention.

Six inputs. Every storm is defined by central pressure CP, radius of maximum wind $R_{\max}$, translation speed VT, the Holland shape parameter B, a gradient-to-surface conversion factor CF, and far-field pressure FFP (pressure deficit $\Delta p = \mathrm{FFP} - \mathrm{CP}$).

One population of 200, sampled with real dependence. Rather than 100 storms in each of Saffir–Simpson categories 1, 3, and 5 drawn independently, the app runs a single **constrained lumped design of 200 storms** whose inputs honour the dependence structure of the HURDAT2 best-track archive—most importantly the central-pressure/radius coupling ($\mathrm{CP} \sim R_{\max}$, correlation $r \approx 0.43$) and a radius-dependent shape parameter (B–$R_{\max}$, via Eq. 7 of Powell et al. [5]). Here

B is a *direct* design input, not a quantile mapped through a distribution. The legacy 3-category design is retained as an independent-inputs baseline and cross-check.

Why it matters. Fixing a Saffir–Simpson category nearly fixes CP, so within a category CP cannot express its influence—the classic framing structurally hides the most important variable. Lumping the categories into one realistically-sampled population lets CP vary over its full range, and it immediately becomes the leading driver (see the Statistics section).

4 Vulnerability: from wind to damage

Damage is a **Mean Damage Ratio** (MDR), the fraction of a structure's value lost, following the HAZUS/ARA wind-vulnerability methodology [12, 13, 14], given as a smooth logistic function of peak wind: near zero below tropical-storm force, rising steeply through the 100–150 mph range, and saturating toward 1 for the most extreme winds. The curve's midpoint, steepness, and cap are adjustable in the viewer, so the vulnerability leg is itself a swappable model component. Because MDR is a ratio, it is exposure-agnostic: it says what *fraction* is lost, and the exposure book (next section) says of *how much*.

5 Actuarial: exposure, loss cost, and financial metrics

Loss cost. At a land vertex the ground-up loss is $\mathrm{LC} = \mathrm{MDR}(\mathrm{wind}) \times \mathrm{exposure}$. The total loss a storm inflicts is $\mathrm{TLC}(i) = \sum_{\mathrm{land}} \mathrm{LC}$, and the headline ratio is $\%\mathrm{TLC}(i) = \mathrm{TLC}(i)/(\text{total exposure})$.

Three exposure books. Exposure is swappable: a **uniform** \$100k/vertex reference (total \$68.2 M), or realistic value surfaces from **Census** (ACS home value) and **Tax Roll** (Florida cadastral) data. The realistic books are extraordinarily concentrated—the densest Miami cell carries some 1500× the median, and the coastal Gini is ≈ 0.84—which changes loss magnitudes by orders of magnitude and, crucially, changes which wind-field details matter (a coastal wind shift moves the loss far more than an inland one).

Financial layer. A financial panel adds the actuarial leg on top of hazard × vulnerability: per-location deductible and limit, an exceedance-probability (EP/OEP) curve of loss versus return period, and metrics—average annual loss (AAL), 50/100/250-year losses, and tail value-at-risk—driven by a single editable annual landfall rate for the lumped population.

6 Statistics: what drives the loss, and how sure are we

The app reports three complementary sensitivity measures over the 200-storm population, following the Commission's sensitivity/uncertainty demonstration [2].

SRC and EPR. The **standardized regression coefficient** (SRC) gives each input's first-order, linear influence (sign and magnitude); the **uncertainty** measure EPR is reported as the variance share SRC_i^2.

Shapley effects — the primary, dependence-aware measure. Because the constrained inputs are correlated *by construction*, the classical Sobol' variance decomposition [16] (which assumes independent inputs) is not strictly valid. The **Shapley effect** [19, 20] borrows the Shapley value from cooperative game theory to apportion output variance among dependent inputs: the shares are non-negative and sum exactly to one whatever the correlations. It is estimated with a given-data nearest-neighbour estimator [21], and is the tool's primary sensitivity method (Figure 2).

Sobol' as an independent-input cross-check. The classical Sobol' indices are still computed—sampling the marginals independently, with the Saltelli and Jansen estimators [17, 18] on the emulator [15]—as a reference. **The two methods agree on the leaders**: on both peak wind and loss cost, **CP dominates** (Shapley share ≈ 0.45–0.51) with $R_{\max}$ second (≈ 0.21–0.29). They diverge where dependence and the decay process bite hardest—on translation speed VT (which the decay makes matter) and on B (coupled to $R_{\max}$)—which is exactly the behaviour a dependence-aware method is needed to get right. Agreement on the headline is the reassurance of running two independent methods: *we did the irregular (Shapley) and the regular (Sobol') analysis, and they concur that central pressure leads.*

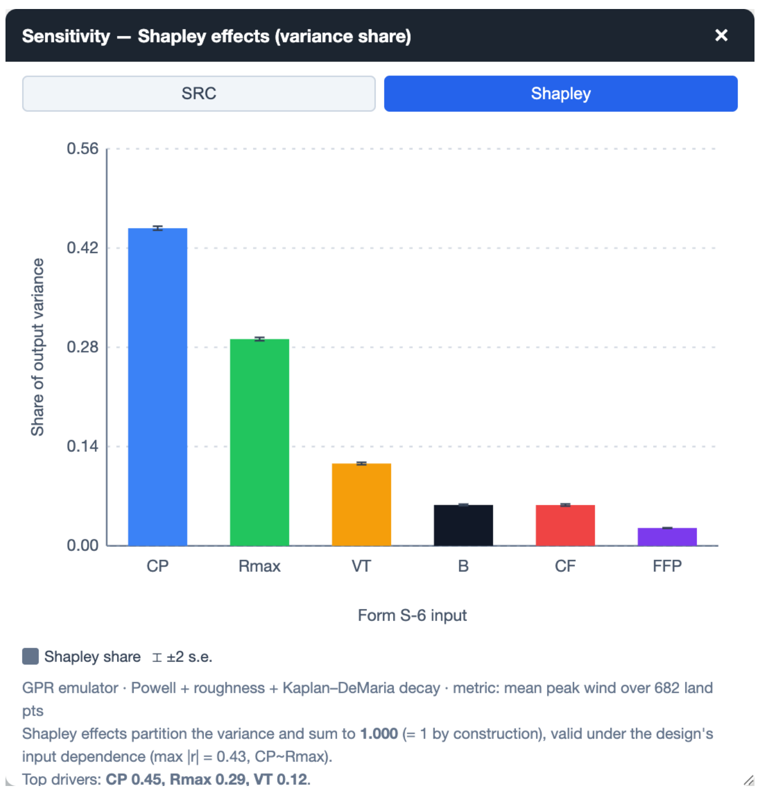

Figure 2: Shapley effects for peak wind on the constrained 200-storm population (shipped default land configuration). CP dominates; $R_{\max}$ second; the six shares sum to one by construction. This overturns the legacy per-category ranking, in which storm size or profile shape led, because lumping frees CP to vary.

Metamodels and the profiler. The sensitivity measures run on a **metamodel** (emulator) fit to the 200 runs—a second-order response surface, a Gaussian-process regression, and a neural network, which agree and (because the simulator is smooth) reach $R^2 \approx 1$; their value is diagnostic, not predictive. An interactive **profiler** and **interaction matrix** let the user bend one input at a time and see the response and the pairwise interactions directly. A hold-out test on unseen storms confirms the emulator predicts to $R^2 \geq 0.998$.

Spatial sensitivity. Sensitivity is also mapped *per vertex*: the dominant input varies across the grid, so a single grid-averaged ranking can mislead—an off-track vertex is governed by storm size (does the wind reach it at all?) while the domain average is governed by central pressure.

7 Computer science: the implementation

The deployed application is a **zero-backend static web page** (HTML/CSS/JavaScript + Leaflet); it can be served from any static host. The pattern throughout is **precompute-then-evaluate**: expensive physics (the Powell PDE, the machine-learning emulators, the dynamic-storm animation frames) is computed offline in Python and exported to compact JSON, and the browser only *samples* it—so model switches and analyses are instant and the site needs no server (Figure 3).

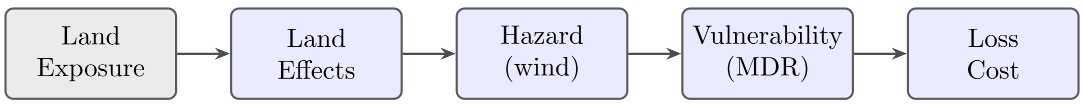

Figure 3: The catastrophe-model chain the app implements. Exposure is conditioned by the land effects, which shape the hazard; the hazard drives vulnerability; and vulnerability against exposure yields loss costs. Each stage is a swappable model.

Interface at a glance. The viewer offers model selection, an input-vector slider over the 200 storms (no category selector—the population is unstratified), the land-effect checkboxes, colour-by modes (wind, loss, land/water, dominant-sensitivity), a points/contour display toggle, light/dark themes, hover read-outs, saved points of interest with printable detail pages, an optional storm animation, and the analysis panels (sensitivity, uncertainty, profiler, loss-cost CDF, and the financial EP curve).

Reproducibility. The pipeline is public-domain and self-contained: a fresh clone rebuilds every wind field, the damage curve, and every figure in the reports. The Powell PDE solver and the wind-vulnerability tool are vendored in-repo; nothing the pipeline needs lives outside the clone.

8 The essentials, in five lines

- **Meteorology:** three wind models (Powell/Holland/Willoughby) plus a time-dependent Powell, over the specified 840-vertex South-Florida grid, with roughness and inland-decay land effects.

- **Design:** one realistically-dependent population of 200 storms, not independent per-category draws.

- **Statistics:** dependence-aware Shapley effects (primary) cross-checked by classical Sobol' (reference)—**central pressure is the dominant driver**, $R_{\max}$ second.

- **Loss:** MDR damage curve × a swappable exposure book (uniform, Census, or Tax Roll), with a full financial exceedance layer.
- **Software:** a reproducible, zero-backend static web app—precompute in Python, evaluate in the browser.

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